Appendix B

Three Fundamental Models of Stock Returns

A pessimist knows the price of everything and the value of nothing.—Oscar Wilde

B.1 THE DIVIDEND DISCOUNT MODEL

The most important concept in finance is the concept of *present discounted value* (PDV). One of the most famous models for valuing securities is based on this simple concept. The idea of present discounted value is that since there is a positive interest rate, a dollar today is worth more than a dollar tomorrow, irrespective of the uncertainty of tomorrow. I can take one dollar today, invest it at the positive interest rate, and end up with more than a dollar tomorrow.

This simple concept can be extended to valuing assets. If I have a paper that guarantees receipt of $8.00 in 6 months and $108 in 1 year, how much should you be willing to pay for that piece of paper? If the interest rate is positive, you should clearly value the paper at less than $116 because if you invest $8 for 6 months and $108 for 1 year, you will have more than $116. In fact, with a flat yield curve, we would value the security as

$$P = \frac{8}{(1 + r/2)} + \frac{108}{(1 + r/2)^2} \tag{B.1}$$

where r represents the annual interest rate on 6-month and 1-year interest-bearing instruments. For example, if $r = 0.03$, or 3%, then $P = 112.71$.

Suppose that we have a more complicated instrument that pays cash flows into the very distant future. Suppose that the security provides annual cash flows for n years. The price of the security is the PDV of the future cash flows.

$$P = CF_1\delta + CF_2\delta^2 + ... + CF_n\delta^n \tag{B.2}$$

where δ equals some positive discount rate, and $\delta < 1$. The discount rate varies with the circumstances and the demands of the investor. Often, $\delta = \frac{1}{1+r}$, which means that all years' cash flows are discounted at the same prevailing interest rate. Discounting at a constant rate assumes a flat term structure of interest rates. Other times, $\delta = \frac{1}{1+r_{CE}}$, where r_{CE} is an investor's *required return on equity*. If the investment is very risky, the investor might require a high r_{CE}. We shall use r to represent the required return on equity throughout the rest of this appendix.

A little bit of mathematical manipulation of this equation will be useful to us later. Firstly, if the cash flows are *constant* throughout the life of the security (that is, if $CF_1 = CF_2 = ... = CF_n = CF$), then Eq. (B.2) becomes[1]

$$P = CF\delta\left(\frac{1-\delta^n}{1-\delta}\right) \tag{B.3}$$

In addition, if the cash flows occur into infinity, then δ^n converges to 0, and thus, as $n \to \infty$, then

$$P = CF\left(\frac{\delta}{1-\delta}\right) \tag{B.4}$$

B.1.1 The Single-Stage DDM

With this basic set of mathematics, we can now derive the *dividend discount model* (DDM). The DDM is a model that prices a firm according to the PDV of its dividend payments because those are all that an investor receives directly from the firm. The DDM is a reasonable way to evaluate companies that pay stable, predictable dividends. Some additional assumptions round out the model. We assume that dividends will grow at a constant rate g forever either because of inflation or because the company just will increase dividend payments over time. We also assume that the company will exist and pay dividends forever; that is, $n = \infty$. Finally, we make the assumption that the rate of return that the investor requires is greater than the growth rate of dividends. Thus $r > g$. These assumptions allow us to use Eq. (B.4) to value this company. The preceding equation is easily transformed so that $CF = D_0$, where D_0 is this year's dividend, $\delta = \frac{1+g}{(1+r)}$, and r is the required return on equity.

$$P = \frac{D_0(1+g)}{r-g} \tag{B.5}$$

Consider an example of the application of the model. Suppose that Mr. Wolfgang Chincarini is the new CEO of Inter-Milan (an Italian soccer team). This year, Inter-Milan's dividend was \$3. The company expects dividends to grow at a rate of 5% for the foreseeable future. The required return on equity is 9%. According to the model, the price of this security should be $P = \frac{D_0(1+g)}{r-g} = \frac{3(1+0.05)}{0.09-0.05} = 78.75$.

This model, known as the single-stage DDM or just DDM, should be used for ranking the relative values of stable, established companies with consistent dividends. For growth companies, the more appropriate model is the *two-stage DDM*, a cousin of the single-stage DDM. In both versions of the DDM, the quality of the inputs is critical to the reliability of the valuation. An analyst may use his or her own forecasts of earnings and dividend growth or forecasts from the IBES database.

[1]This series is derived the following way: Let $S_n = 1 + \delta + \delta^2 + ... + \delta^n$. Then $S_n - \delta S_n = 1 - \delta^{n+1}$, and thus $S_n = \frac{1-\delta^{n+1}}{1-\delta}$.

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B.1.2 The Two-Stage DDM

The two-stage DDM is derived mathematically in a very similar fashion to the single-stage DDM. The analyst assumes that the company in question goes through two stages of growth. The company first enters a stage of supernormal (high) growth and then settles into a stage of normal growth. Let's introduce g_h and g_l to represent the high and low rates of growth in the first and second stages, respectively. Let n_h represent the periods of high growth, and n represent the number of years of the second stage of growth, which we will assume to go on for infinity. The two-stage DDM is solved by breaking the problem into its two stages. First, we find the PDV of the dividends during the supernormal growth period. Then, since after the supernormal growth stage dividends grow at the lower rate forever, we take the value of the dividends as of the end of the supernormal stage and apply the valuation of the single-stage DDM. The present discounted value of the dividends from the supernormal stage is

$$\begin{aligned} PDV_h &= \frac{D_0(1+g_h)}{(1+r)} + \frac{D_0(1+g_h)^2}{(1+r)^2} + ... + \frac{D_0(1+g_h)^{n_h}}{(1+r)^{n_h}} \\ &= \frac{D_0(1+g_h)}{(r-g_h)}\left[1 - \left(\frac{1+g_h}{1+r}\right)^{n_h}\right] \end{aligned} \tag{B.6}$$

The present discounted value of the dividends from the normal, lower-growth stage is

$$PDV_h = \frac{D_0(1+g_h)^{n_h}(1+g_l)}{(1+r)^{n_h}(r-g_l)} \tag{B.7}$$

Thus, the price of the security using the two-stage DDM is

$$P = \frac{D_0(1+g_h)}{(r-g_h)}\left[1 - \left(\frac{1+g_h}{1+r}\right)^{n_h}\right] + \frac{D_0(1+g_h)^{n_h}(1+g_l)}{(1+r)^{n_h}(r-g_l)} \tag{B.8}$$

Consider an example of the application of the model. Suppose that Mr. Angelo Chincarini and Ms. Gemma Chincarini are the new CEOs of the Jetsons Future Company, a growth company that intends to deliver high-quality assistance to all people for engaging in their day-to-day life. Based on your analysis, you believe that the company will have very strong dividend growth for 2 years at 33%, which will subside to 8% after that period into the foreseeable future. The YTM of the company's debt is 16%, and the risk premium of the stock is about 4%. The current dividend is \$2.00. What is the "fair value" of this company's share price? First, let's compute the required return on equity as the company's debt yield plus the risk premium on the stock, thus $r = 0.20$. We can now use our formula to find the fair price. Thus, $P = \frac{D_0(1+g_h)}{(r-g_h)}\left[1 - \left(\frac{1+g_h}{1+r}\right)^{n_h}\right] + \frac{D_0(1+g_h)^{n_h}(1+g_l)}{(1+r)^{n_h}(r-g_l)} = \frac{2(1+0.33)}{(0.20-0.33)}\left[1 - \left(\frac{1+0.33}{1+0.20}\right)^2\right] + \frac{2(1+0.33)^2(1+0.08)}{(1+0.20)^2(0.20-0.08)} = 26.78$. Thus the fair value of the company's share price is \$26.78 according to the two-stage DDM and our assumptions.

Although many quantitative portfolio managers do not use the DDM, some do. One way to use the DDM is to compute the DDM price of a stock and subtract its actual price from the DDM price. This difference will be positive if the stock is undervalued and negative if the stock is overvalued. The portfolio manager can rank all the stocks in the investment universe according to this difference. Stocks with a higher difference are preferred because they are undervalued.

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Another good way to use the DDM model is to calculate implied growth rates from actual stock prices. Rather than try to calculate the right price for a stock, we can find out what the model says other parameters should be, given the actual market prices. The parameter typically found this way is the implied growth rate of a company's earnings. A simple manipulation of the single-stage DDM leads to the following equation for the implied growth rate:

$$g = \frac{Pr - D_0}{D + P} \tag{B.9}$$

Given the stock price, the current dividend, and the required return on equity, one can infer the implied growth assumption of the market. A more common variant is to make the substitution of $D_0 = kE_0$, where k is the payout ratio. This transforms the equation into

$$g = \frac{\frac{P}{E_0}r}{\frac{P}{E_0} + k} \tag{B.10}$$

From this equation, an analyst or portfolio manager can determine what growth rate a certain P/E ratio implies. For example, suppose that the P/E ratio for the Standard and Poor's (S&P) 500 is 28.11 (this can be found on S&Ps web site). Suppose that the dividend yield (D/P) is 1.63%. Thus $k \approx 0.46$. Suppose the 10-year bond yield is 4.28%, there is a 4% risk premium, and the required return on equities is 8.28%. Rearranging the equation, one finds that the implicit growth rate of earnings is 8.15%. An analyst might think this is too high or too low and could take appropriate actions in response.

Although we are not major users of the DDM model, we thought it might be interesting to show how the concepts apply to diverse, realistic situations. Three applications of the DDM are described below.

B.1.3 Application 1: Housing Prices

From 2002 to 2005, there had been constant discussion of the overvaluation of housing prices in the United States. One of the most common arguments for overvaluation was that the home price-to-income ratio ($\frac{P_H}{I}$) is too high. If this ratio is high, it is argued, the cost of a home represents an overly great burden on a family's income. Let's normalize median family income to 100 in 1990. From 1990 to 2003, the median family income in the United States increased by 50%.[2] Over this same period, the Office of Federal Housing Enterprise Oversight (OFHEO) index of housing prices increased by 72%. If we also normalize the housing price index to 100 in 1990, then (P_H/I) increased from 1 to 1.15, or 15%, between 1990 and 2003. This implies that in comparison with their situation in 1990, home owners were overextended owing to high housing prices in 2003. But are we missing something?

McCarthy and Peach (2005) claim that we should use standard underwriting criteria for mortgages rather than housing prices as the basis for determining housing affordability. In other words, we ought to consider a cash-flow-to-income (CF/I) valuation of housing prices, where the cashflow is the 30-year annual mortgage payment that would be required for a given loan to cover the cost of the house. Thus, to figure out the yearly payment on a 30-year mortgage, we can use a variant of our cash-flow discount model. From Eq. (B.3), we have

[2] See McCarthy and Peach (2005).

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$$\begin{aligned} P &= CF\delta\left(\frac{1-\delta^n}{1-\delta}\right) \\ \rightarrow CF &= \frac{P}{\delta\left(\frac{1-\delta^n}{1-\delta}\right)} \end{aligned} \tag{B.11}$$

This formula gives the required mortgage payment for a given house price. In 1990 the average nominal interest rate on 30-year mortgages was about 10%, whereas in 2003 it was about 5.75%. Plugging these values into the formula, we find that the cash flow required on a house that cost 100 (in 1990) and 172 (in 2003) was $10.61 in 1990 and $12.16 in 2003. The CF/I ratio in 1990 was 0.106, and in 2003 it was 0.081. In terms of cash flow drain on a person's annual income, therefore, the ratio actually had decreased. This might lead us to the opposite of the popular conclusion: Housing prices in fact may be undervalued.[3]

B.1.4 Application 2: The Lockheed-Martin Award

On Friday, October 26, 2001, it was announced that the Department of Defense had awarded Lockheed-Martin Corporation a $200 billion contract to build its new radar-evading Joint Strike Fighter. Lockheed beat out a major competitor, Boeing, for the contract. A trader might wonder how much Lockheed's (ticker symbol LMT) price should increase owing to this contract. If he or she gets a good handle on the price increase, he or she might be able to make money from a market over- or under-reaction. A variant of the DDM model can be very useful at this stage. Let's take Eq. (B.3), substitute earnings for cash flows, and look at how the change in earnings will affect the change in the stock price. With the substitution,

$$\Delta P = \Delta E\delta\left(\frac{1-\delta^n}{1-\delta}\right) \tag{B.12}$$

where ΔP represents the change in price of LMT owing to the increasing earnings of the company for the contract, ΔE represents that change in earnings over time due to the contract, and δ is our discount factor, defined as $\delta = \frac{1}{1+r}$.

The values of the inputs are critical at this stage. At the time of the announcement, analysts who followed the company closely made the following observations: (1) The contract would not affect Lockheed's bottom line in the short run, (2) the contract extended over 30 years, (3) between 25% and 30% of the contract value would go to other participants, and (4) Boeing was Lockheed's one major competitor for the contract.

Although there are many ways to construct the inputs, we will focus on one method for illustration purposes. After 30% of the $200 billion contract value goes to other participants, Lockheed is left with $140 billion. We divide this amount by the approximately 440.52 million shares outstanding at the end of October 2001 to obtain the revenue effect of the contract, $317.81 in revenue per share. We then take Lockheed's historical net profit margin of 2.7% and apply that to this value to obtain an earnings per share (EPS) for the contract of $8.58. We divide the contract's EPS by 30

[3] We ignored tax effects in these examples. A more sophisticated analysis would consider the after-tax cost of housing. Higher housing prices raise the tax due to holders of property, but this might be offset by deductions from mortgage interest and property tax.

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for the assumed 30 years over which these revenues would be distributed. This leaves us with a net annual EPS of \$0.286 for the contract. At this point we can apply the DDM model. The change in EPS from the contract is 0.286, $n = 30$, and we assume that the required return on equity is 6.5% for LMT (which has a very low β). If we plug these values into Eq. (B.12), we obtain a value of \$3.73. Thus, if we believe our assumptions, the price of LMT should increase by \$3.73 on the news.

We still have to take into consideration, however, the expectations that the market had prior to the awarding of the contract. The fourth piece of information we have about the contract is that Lockheed only faced competition from one other company, Boeing. The market would have driven up LMT's price prior to the announcement of the contract on the expectation that Lockheed had a good chance of winning the contract. Even if the market had assumed that the contract was equally likely to go to either Boeing or Lockheed, investors probably would have already driven up LMT's price by about \$1.87, about 50% of the \$3.73 per share value of the contract. Assuming that this was the case, we would only expect another \$1.87 increase in LMT's price after the announcement.

Before the announcement of the contract, LMT was trading at \$49.62 per share. Based on our DDM analysis, we would have expected the price to go up \$1.87 to \$51.49 the Monday following the announcement. We therefore would have advised the trader to short LMT if it opened above \$51.49 or buy it if it opened below \$51.49.[4] Of course, the context of the September 11 attacks would have raised additional considerations regarding a stock such as LMT in October 2001, but the DDM model would have provided these guidelines for trading purely on the news of the contract.

B.1.5 Application 3: The Fed Model

The Fed model is a very popular model that relates the earnings yield of the stock market to the bond yield.[5] According to the model, the market is in equilibrium when these yields are similar to each other. The earnings "yield" of the stock market is given by the inverse of the P/E ratio. (Although a stock's yield is really its dividend, the stock's earnings also can be distributed to shareholders eventually.) For bonds, the yield is the comparable return on the bond asset class. The idea is that there should be some relationship between the two types of yields in equilibrium. If interest rates decrease, for instance, stock prices may go up, resulting in higher P/E ratios and lower earnings yields in equilibrium.

The model divides the earnings yield by the bond yield. When this ratio is greater than 1, the stock market is considered *undervalued*; when the ratio is less than 1, the stock market is considered *overvalued*. To come up with the earnings yield, one divides the 12-month forward consensus earnings of the S&P 500 (E) by the value of the S&P 500 Index (P). The next step is to divide the E/P ratio of the S&P 500 by the yield on 10-year U.S. Treasury bonds. For example, suppose that the 10-year bond yield is 4.31%. Suppose that the S&P 500 P/E ratio is 42.54 (thus the earnings yield is about 2.3%). The ratio of E/P to the 10-year yield is equal to 0.55. According to the Fed model, the market is overvalued.

The Fed model is used widely by practitioners to determine whether the stock market is over- or undervalued, but in our opinion, it is a poor model. Accurate historical backtests of the Fed model have not fared too well partly because of the unrealistic assumptions that drive it. The

[4] On Monday, October 29, 2001, LMT in fact opened at \$52.98 and closed at \$49.00.

[5] The model takes its name from the Federal Reserve's 1997 Monetary Policy Report, although it probably was not the intention of the Fed to popularize an investment model as such.

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Fed model merely highlights the strengths of the DDM, which we can use to understand the Fed model's failings.[6] Let's start with the basic DDM model and find an equilibrium formula for the P/E ratio:

$$P = \frac{D_0(1+g)}{r-g} \tag{B.13}$$

We substitute for dividends kE, where k is the payout ratio (the percentage of earnings paid out as dividends), and E_0 represents earnings today. Let's also call E_1 earnings next year, which we typically would calculate as $E_0(1+g)$ but in the Fed model are set equal to the analyst consensus figure. A little manipulation of the preceding equation reveals that

$$\frac{P}{E_1} = \frac{k}{r-g} \tag{B.14}$$

We also can also set the growth rate of earnings equal to $(1-k)ROE$. The equation then becomes:

$$\frac{P}{E_1} = \frac{k}{r-(1-k)ROE} \tag{B.15}$$

If we assume that the required return on equity (r) equals the 10-year treasury yield (that is, $r = y_{10}$), and that the return-on-equity equals the 10-year Treasury yield (that is, $ROE = y_{10}$), then the equation reduces to $\frac{P}{E_1} = \frac{1}{y_{10}}$. Now the problems inherent in the Fed model become clear. The ROE of the S&P 500 historically always has been much higher than the 10-year Treasury yield. Also, the movements of the index have not been related to the 10-year Treasury; since 1977, the correlation between them has been -0.42. Finally, the required return-on-equity should not, theoretically, be equal to that of a government security. Rather, it should be related to the risk on equities, which is something that changes frequently.

B.2 THE CAPITAL ASSET PRICING MODEL

One of the most famous models of expected stock returns is the capital asset pricing model (CAPM). The model describes stock returns and risks in equilibrium based on a set of assumptions about investor behavior. Several authors contributed to what is now known as the CAPM. The model originated as a description of the decisions of Markowitz-type investors, investors who seek to maximize utility given capital market opportunities. Further assumptions were made as the model evolved into a formal theory. In particular, it was assumed that all investors could borrow and lend at the risk-free rate, that there were no taxes or transactions costs, that all investors had the same time horizon, and that all investors had identical expectations for asset returns. Given these assumptions, the CAPM says that all investors will hold some portion of the Markowitz-efficient portfolio and some portion at the risk-free rate. Each investor's relative appetite for risk will determine these portions. The sum of all investments equals the total wealth in the market, so the Markowitz-efficient portfolio is the *market portfolio*, or world portfolio. Since (according to Sharpe

[6] See Asness (2003) for additional problems with the Fed model.

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and others) the market portfolio is the only risky asset relevant to investors' decisions, individual stocks' returns depend only on their levels of exposure to the market portfolio and on the market portfolio's expected return. The CAPM describes this relationship in the form of the *security market line*, which translate's a risky asset's β, or exposure to the market, into its expected return. This relationship is also known as the *single-factor model*. More formally,

$$E(r_i) - r_f = \beta\,[E(r_m) - r_f] \tag{B.16}$$

where $\beta = \frac{C(r_i, r_m)}{V(r_m)}$.[7]

The CAPM is important in many ways for quantitative equity portfolio management (QEPM). First, it is one of the ways in which α is defined, as explained in Chapter 2. Second, it is a method for controlling risk, including in portfolio optimization and for constructing market-neutral portfolios. Third, CAPM concepts are used in performance measurement to determine a quantitative portfolio manager's risk-adjusted returns.

B.3 THE ARBITRAGE PRICING THEORY MODEL

Although still widely taught and used, the CAPM has a number of well-recognized limitations as a model for security pricing. It is unable to explain many market anomalies,[8] and some of its stringent assumptions are downright unrealistic. In 1976, Stephen Ross developed the main alternate to the CAPM, the arbitrage pricing theory (APT). The models of stock returns in this book (particularly the factor models of Chapters 6 and 7), most of the risk and return models built by quantitative managers, and the models created by leading software providers are all based on APT.

Unlike the CAPM, which says that every security's return depends on only the performance of the market, the APT posits that security returns (and risk premia) are best explained by more than one factor. The first postulate of APT is that security returns can be represented as a linear combination of factor risks. Thus

$$\begin{aligned} r_i - E(r_{i,t}) &= \beta_{i,1} f_{1,t} + ... + \beta_{i,K} f_{K,t} + \epsilon_{i,t} \\ E(\epsilon_i f) = 0,\ E(\epsilon_i^2) = \sigma_i^2,\ E(f_{k,t}) &= 0, E(\epsilon_{i,t}\epsilon_{i,t+j}) = 0 \quad \forall j > 0 \end{aligned} \tag{B.18}$$

where $r_{i,t}$ is the total return on asset i for period t, $E(r_{i,t})$ is the expected return on asset i at the beginning of period t, β_{ik} is the risk exposure (a.k.a. *factor loading* or *factor sensitivity*) of asset i to factor k, $f_{k,t}$ is the realization of factor k at the end of time period t, and $\epsilon_{i,t}$ is the specific shock

[7] One also should be aware of Black's version of the CAPM, known as the *zero-β CAPM* (Black, 1972). Black considered the problem of not having a risk-free asset available. The expected return of an asset in excess of the zero-β return is linearly related to its β. The equation is:

$$E(r_i) - E(r_{0m}) = \beta[E(r_m) - E(r_{0m})] \tag{B.17}$$

where r_m is the return on the market portfolio, and r_{0m} is the return on the zero-β portfolio associated with m. It is actually the portfolio with the minimum variance of all portfolios uncorrelated with the market. The Black model is defined in inflation-adjusted returns or real returns. Estimation of this model is naturally more complex because the zero-β asset is unobserved. There are, however, econometric methods to test for the model.

[8] See Chapter 2.

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to asset i during period t. Note that $f_{k,t}$ is the surprise component of the factor (i.e., factor shock). It's mean is equal to 0. Note also that asset-specific shocks do not affect the expected stock return. Finally, it is assumed that the factor realizations and asset-specific shocks are uncorrelated.

The second and very important postulate of the APT is that *pure arbitrage* opportunities do not exist. An investor cannot earn a positive expected rate of return in any combination of assets without taking some *risk*. This powerful postulate makes the APT independent of investor preferences and probability distributions. It also provides a fascinating insight into the tradeoffs between risk and return.

Given additional assumptions,[9] the APT's first and second postulates can be combined as

$$E(r_{i,t}) = P_0 + \beta_{i,1}P_{1,t} + ... + \beta_{i,K}P_{K,t}. \tag{B.19}$$

where the P_ks represent the price of risk or the risk premium for the kth risk factor. This equation is monumental because it states that the expected return of an asset is exactly reflected by its exposure to different types of economy-wide risk. The P_0 risk factor has a special meaning. Suppose that we had a portfolio so diversified that residual risk equaled zero (that is, $\sum_{i=1}^{N} w_{i,t}\epsilon_{i,t} = 0$) and that the factor exposures were zero (that is, $\sum_{i=1}^{N} \beta_{i,k} = 0 \; \forall k = 1, ...K$). Such a portfolio would have zero risk, and its expected return would equal P_0. Thus P_0 must equal the risk-free rate. Similarly, P_k is the excess return over the risk-free rate of a portfolio that has one unit of exposure to factor k and zero exposure to all other factors. The full APT model is obtained by combining the equations into:

$$\begin{aligned} r_i - P_0 &= \beta_{i,1}(P_1 + f_{1,t}) + ... + \beta_{i,K}(P_K + f_{K,t}) + \epsilon_{i,t} \\ E(\epsilon_i f) = 0,\; E(\epsilon_i^2) = \sigma_i^2,\; E(f_{k,t}) = 0,\; E(\epsilon_{i,t}\epsilon_{i,t+j}) &= 0 \quad \forall j > 0 \end{aligned} \tag{B.20}$$

One of the drawbacks of the APT is that it is a mathematical theory, and unfortunately, the theory does not tell us which factors should go into the multifactor model. Practitioners and academics search continuously for risk factors that adequately explain security returns. There are two broad theoretical approaches to assigning factors, which we discussed in detail in Chapters 6 and 7.

One approach is the macroeconomic approach (see Chapter 7), which posits that security returns must be related to underlying, systematic risks in the economy, such as industrial production growth and unexpected inflation. This is the model that Stephen Ross must have had in mind when he derived the theory, since all his empirical papers on the subject use macroeconomic factors.

Another approach is the microeconomic or fundamental approach (see Chapter 6), which bases the factor model on characteristics of the company, such as the P/B ratio. The macroeconomic approach is more solidly grounded in theory because one would imagine that systematic risk factors are more related to economic forces that affect the entire economy. However, the microeconomic or fundamental factor model is used more widely in practice. There are probably several reasons for its prevalence. Characteristics of firms may be easier for people to understand than economic

[9]For a discussion of additional assumptions needed for the equation to be exact, rather than approximate, see Connor (1984). Dybvig (1985) and Grinblatt and Titman (1985) show that the deviations from exact factor pricing are likely to be negligible, and so the exact pricing relation is justified. Merton's (1973) ICAPM combined additional assumptions, also producing an exact factor pricing relation.

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indicators. The fundamental factor model is also easier to estimate than the macroeconomic factor model is. The fundamental factor model was publicized by the academic research of Eugene Fama and Kenneth French through a series of publications (Fama and French (1992,1993)) as well as popularized among practitioners by the success of software produced for portfolio managers, including Barra and FactSet. Finally, the firm-characteristic model predicts better out-of-sample than does its macroeconomic counterpart.[10]

[10]This is one of the reasons Eugene Fama gives for using the model, despite its theoretical inferiority to the macroeconomic model.

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